

## Probe Arm

### F2.13: Probe Arm Device Display

08:44:04

N2

Current Commands

Devices Timers Macro Vars Active Codes ATM Calculator Media

Mechanisms

| Device        | State     |
|---------------|-----------|
| Main Spindle  | Off       |
| Parts Catcher | Retracted |
| Probe Arm     | Retracted |

**Probe Arm**

**F2** Extend

\*\*Check that the probe arm has room to extend, otherwise you may damage it.

\*\*Use [F2] to extend the arm for probing or retract it out of the way for continued operation.

The **Probe Arm** option in **Devices** allows you to **Extend** and **Retract** the probe arm. The door must be fully opened or fully closed.

- Use the cursor arrow keys to move from field to field.
- Press **[F2]** to extend the probe arm and press **[F2]** to retract the probe arm.

## Bar Feeder

### F2.14: Bar Feeder Setup Display

10:19:09

N0

Current Commands

Devices Timers Macro Vars Active Codes ATM Calculator Media

Mechanisms Bar Feeder



**F2** Load and Measure Bar

**F3** Advance Bar

**F4** Set Collet Face Position

**INSERT** Set Push Rod Offset

Bar Feeder System Variables

| Description                           | Value   | Unit |
|---------------------------------------|---------|------|
| Length of Longest Bar                 | 48.0000 | IN   |
| Total Push Length (D)                 | 0.0000  | IN   |
| Total Initial Push Length (F)         | 0.0000  | IN   |
| Minimum Clamping Length (G)           | 0.0000  | IN   |
| Maximum Number of Parts               | 0       |      |
| Maximum Number of Bars                | 0       |      |
| Set up 1: Load Bar and Measure        | --      |      |
| Set up 2: Adjust Transfer Tray Height | --      |      |

The **Bar Feeder** tab on **Devices** allows you to set up the Bar feeder system variables.

- Use the cursor arrow keys to move from field to field.

## Time Adjustment

Follow this procedure to adjust the date or time.

1. Select the **Timers** page in Current Commands.
2. Use the cursor arrow keys to highlight the **Date:**, **Time:**, or **Time Zone** field.
3. Press **[EMERGENCY STOP]**.
4. In the **Date:** field, type the new date in the format **MM-DD-YYYY**, including the hyphens.
5. In the **Time:** field, type the new time in the format **HH:MM**, including the colon. Press **[SHIFT]** and then **[9]** to type the colon.

6. In the **Time Zone:** field, press ENTER to select from the list of time zones. You can type search terms in the pop-up window to narrow the list. For example, type **PST** to find Pacific Standard Time. Highlight the time zone you want to use.
7. Press **[ENTER]**.

## Timer and Counter Reset

You can reset the power-on, cycle-start, and feed cutting timers. You can also reset the M30 counters.

1. Select the **Timers** page in Current Commands.
2. Use the cursor arrow keys to highlight the name of the timer or counter that you want to reset.
3. Press **[ORIGIN]** to reset the timer or counter.



### **TIP:**

*You can reset the M30 counters independently to track finished parts in two different ways; for example, parts finished in a shift and total parts finished.*

## Current Commands - Active Codes

**F2.15:** Active Codes Display Example

| Current Commands |               |            |              |     |                          |       |
|------------------|---------------|------------|--------------|-----|--------------------------|-------|
| Devices          | Timers        | Macro Vars | Active Codes | ATM | Calculator               | Media |
| G-Codes          | Address Codes |            | DHMT Codes   |     | Speeds & Feeds           |       |
| G00              | N             | 0          | D            | 00  | Programmed Feed Rate     | 0.    |
| G18              | X             | 0.         | H            | 00  | Actual Feed Rate         | 0.    |
| G90              | Y             | 0.         | M            | 00  | Programmed Spindle Speed | 0.    |
| G113             | Z             | 0.         | T            | 00  | Commanded Spindle Speed  | 0.    |
| G20              | I             | 0.         |              |     | Actual Spindle Speed     | 0.    |
| G40              | J             | 0.         |              |     | Coolant Spigot Position  |       |
| G49              | K             | 0.         |              |     |                          |       |
| G80              | P             | 0          |              |     |                          |       |
| G99              | Q             | 0.         |              |     |                          |       |
| G50              | R             | 0.         |              |     |                          |       |
| G54              | O             | 000000     |              |     |                          |       |
| G97              | A             | 0.         |              |     |                          |       |
| G64              | B             | 0.         |              |     |                          |       |
| G69              | C             | 0.         |              |     |                          |       |
|                  | U             | 0.         |              |     |                          |       |
|                  | V             | 0.         |              |     |                          |       |
|                  | W             | 0.         |              |     |                          |       |
|                  | E             | 0.         |              |     |                          |       |

This display gives read-only, real-time information about the codes that are currently active in the program; specifically, the codes that define the current motion type (rapid vs linear feed vs circular feed), positioning system (absolute vs incremental), cutter compensation (left, right or off), active canned cycle, and work offset. This display also gives the active Dnn, Hnn, Tnn, and most recent M-code. If an alarm is active, this shows a quick display of the active alarm instead of the active codes.

### Calculator

The calculator tab includes calculators for basic mathematical functions, milling, and tapping.

- Select the calculator tab in the **[CURRENT COMMANDS]** menu.
- Select the calculator tab that you want to use: **Standard**, **Milling**, or **Tapping**.

## Standard Calculator

### F2.16: Standard Calculator Display

|                |   |   |       |          |        |
|----------------|---|---|-------|----------|--------|
|                |   |   |       |          |        |
| 7              | 8 | 9 | + [D] | +/- [E]  | MS [S] |
| 4              | 5 | 6 | - [J] | sqrt [K] | MR [R] |
| 1              | 2 | 3 | * [P] | % [Q]    | MC [C] |
| (              | 0 | ) | / [V] | Enter    |        |
| Clear [ORIGIN] |   |   | .     |          |        |

**F2** Switch Entry To Input Line

**INSERT** To append to INPUT line.

**ALTER** To replace INPUT line.

**ORIGIN** Reset Calculators

The standard calculator has functions like a simple desktop calculator; with available operations such as addition, subtraction, multiplication, and division, as well as square root and percentage. The calculator lets you easily transfer operations and results to the input line so that you can put them into programs. You can also transfer results into the Milling and Tapping calculators.

- Use the number keys to type operands into the calculator.
- To insert an arithmetic operator, use the letter key that appears in brackets next to the operator you want to insert. These keys are:

| Key | Function            | Key | Function           |
|-----|---------------------|-----|--------------------|
| [D] | Add                 | [K] | Square Root        |
| [J] | Subtract            | [Q] | Percentage         |
| [P] | Multiply            | [S] | Memory Store (MS)  |
| [V] | Divide              | [R] | Memory Recall (MR) |
| [E] | Toggle sign (+ / -) | [C] | Memory Clear (MC)  |

- After you have entered data into the calculator input field, you can do any of the following:



**NOTE:** *These options are available for all calculators.*

Press **[ENTER]** to return the result of your calculation.

Press **[INSERT]** to append the data or the result to the end of the input line.

Press **[ALTER]** to move the data or the result to the input line. This overwrites the current contents of the input line.

Press **[ORIGIN]** to reset the calculator.

Keep the data or the result in the calculator input field and select a different calculator tab. The data in the calculator input field remains available to transfer into the other calculators.

### Milling/Turning Calculator

#### F2.17: Milling/turning Calculator Display

|                 |                                         |        |                                                                                                                                                                                                                   |
|-----------------|-----------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cutter Diameter | <input type="text" value="****, ****"/> | in     | <div>F2</div> Switch Entry To Input Line<br><div>INSERT</div> To append to INPUT line.<br><div>ALTER</div> To replace INPUT line.<br><div>DELETE</div> Clear current input<br><div>ORIGIN</div> Reset Calculators |
| Surface Speed   | <input type="text" value="****, ****"/> | ft/min |                                                                                                                                                                                                                   |
| RPM             | <input type="text" value="****, ****"/> |        |                                                                                                                                                                                                                   |
| Flutes          | <input type="text" value="****, ****"/> |        |                                                                                                                                                                                                                   |
| Feed            | <input type="text" value="****, ****"/> | in/min |                                                                                                                                                                                                                   |
| Chip Load       | <input type="text" value="****, ****"/> | in/tth |                                                                                                                                                                                                                   |
| Work Material   | ◀ ▶ No Material Selected                |        |                                                                                                                                                                                                                   |
| Tool Material   | ◀ ▶ Please Select Work Material         |        |                                                                                                                                                                                                                   |
| Cut Width       | <input type="text" value="****, ****"/> | in     | <div>F3</div> Copy Value From Standard Calculator                                                                                                                                                                 |
| Cut Depth       | <input type="text" value="****, ****"/> | in     | <div>F4</div> Paste Current Value To Standard Calculator                                                                                                                                                          |

Enter a value from 0 - 1000.0000

\* Next to Field Name Denotes Calculated Value

The milling/turning calculator lets you automatically calculate machining parameters based on given information. When you have entered enough information, the calculator automatically displays results in the relevant fields. These fields are marked with an asterisk ( \* ).

- Use the cursor arrow keys to move from field to field.
- Type known values in the appropriate fields. You can also press **[F3]** to copy a value from the standard calculator.
- In the Work Material and Tool Material fields, use the LEFT and RIGHT cursor arrow keys to choose from the available options.
- Calculated values appear highlighted in yellow when they are outside of the recommended range for the workpiece and tool material. Also, when all of the calculator fields contain data (calculated or entered), the milling calculator displays the recommended power for the operation.

### Tapping Calculator

#### F2.18: Tapping Calculator Display

|  |  |               |                            |
|--|--|---------------|----------------------------|
|  |  | <b>F2</b>     | Switch Entry To Input Line |
|  |  | <b>INSERT</b> | To append to INPUT line.   |
|  |  | <b>ALTER</b>  | To replace INPUT line.     |
|  |  | <b>DELETE</b> | Clear current input        |
|  |  | <b>ORIGIN</b> | Reset Calculators          |

|             |                                         |        |
|-------------|-----------------------------------------|--------|
| <b>TPI</b>  | <input type="text"/>                    | rev/in |
| Metric Lead | <input type="text" value="****, ****"/> | mm/rev |
| RPM         | <input type="text" value="****, ****"/> |        |
| Feed        | <input type="text" value="****, ****"/> | in/min |

\* Next to Field Name Denotes Calculated Value

|           |                                            |
|-----------|--------------------------------------------|
| <b>F3</b> | Copy Value From Standard Calculator        |
| <b>F4</b> | Paste Current Value To Standard Calculator |

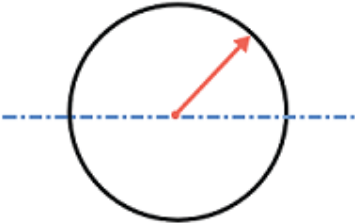
The tapping calculator lets you automatically calculate tapping parameters based on given information. When you have entered enough information, the calculator automatically displays results in the relevant fields. These fields are marked with an asterisk ( \* ).

- Use the cursor arrow keys to move from field to field.

- Type known values in the appropriate fields. You can also press **[F3]** to copy a value from the standard calculator.
- When the calculator has enough information, it puts calculated values in the appropriate fields.

## Arc Calculator

**F2.19:** Arc Calculator Display



**F2** Switch Entry To Input Line

**INSERT** To append to INPUT line.

**ALTER** To replace INPUT line.

**DELETE** Clear current input

**ORIGIN** Reset Calculators

**F3** Open Shape Creator

**F4** Paste Value From Input Line

|              |            |  |    |
|--------------|------------|--|----|
| Radius       | ****, **** |  | in |
| End X        | ****, **** |  | in |
| Arc Center Z | ****, **** |  | in |
| Arc Center X | ****, **** |  | in |
| Z1           | ****, **** |  | in |
| Z2           | ****, **** |  | in |

\* Next to Field Name Denotes Calculated Value

The arc calculator lets you automatically find the start and end points for a radius.

- Use the cursor arrow keys to move from field to field.
- Type known values in the appropriate fields. You can also press **[F3]** to copy a value from the standard calculator.
- When the calculator has enough information, it puts calculated values in the appropriate fields.

## Media Display

M130 Lets you display video with audio and still images during program execution. Some examples of how you can use this feature are:

- Providing visual cues or work instructions during program operation
- Providing images to aid part inspection at certain points in a program
- Demonstrating procedures with video

The correct command format is **M130** (`file.xxx`), where `file.xxx` is the name of the file, plus the path, if necessary. You can also add a second comment in parentheses to appear as a comment at the top of the media window.



**NOTE:**

**M130** uses the subprogram search settings, Settings 251 and 252 in the same way that **M98** does. You can also use the **Insert Media File** command in the editor to easily insert an **M130** code that includes the file path. Refer to page **122** for more information.

**T2.18:** Permitted media file formats

| Standard      | Profile     | Resolution       | Bitrate |
|---------------|-------------|------------------|---------|
| MPEG-2        | Main-High   | 1080 i/p, 30 fps | 50 Mbps |
| MPEG-4 / XviD | SP/ASP      | 1080 i/p, 30 fps | 40 Mbps |
| H.263         | P0/P3       | 16 CIF, 30fps    | 50 Mbps |
| DivX          | 3/4/5/6     | 1080 i/p, 30fps  | 40 Mbps |
| Baseline      | 8192 x 8192 | 120 Mpixel/sec   | -       |
| PNG           | -           | -                | -       |
| JPEG          | -           | -                | -       |



**NOTE:**

*For the fastest loading times, use files with pixel dimensions divisible by 8 (most unedited digital images have these dimensions by default), and a maximum resolution of 1920 x 1080.*

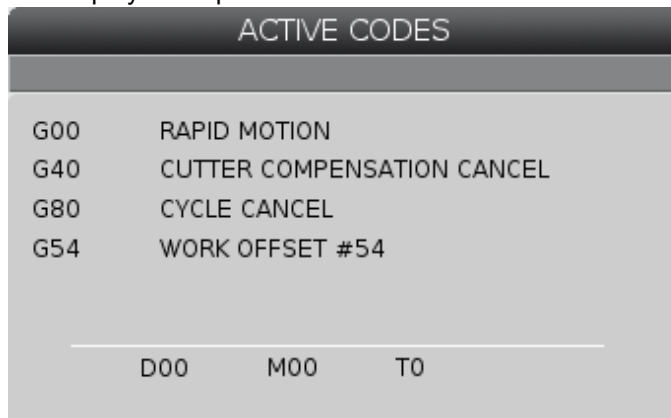
Your media appears in the Media tab under Current Commands. The media displays until the next **M130** displays a different file, or **M131** clears the media tab contents.

## F2.20: Media Display Example - Work Video Instruction during a Program



## Active Codes

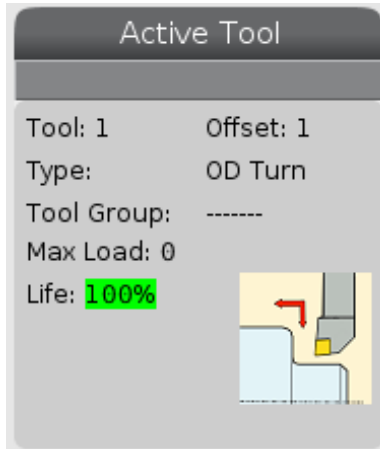
### F2.21: Active Codes Display Example



This display gives read-only, real-time information about the codes that are currently active in the program; specifically, the codes that define the current motion type (rapid vs linear feed vs circular feed), positioning system (absolute vs incremental), cutter compensation (left, right or off), active canned cycle, and work offset. This display also gives the active  $D_{nn}$ ,  $H_{nn}$ ,  $T_{nn}$ , and most recent M-code. If an alarm is active, this shows a quick display of the active alarm instead of the active codes.

## Active Tool

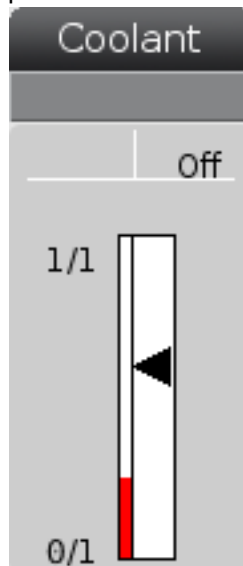
### F2.22: Active Tool Display Example



- The tool number
- The offset number
- The type of tool (if specified in the tool offsets table)
- The tool group number (if specified in the ATM table)
- Maximum tool load (the highest load, in percent, that has been put on the tool)
- The remaining percentage of tool life or tool group
- An example image of the tool type (if specified)

## Coolant Display

### F2.23: Coolant Level Display Example



The coolant display appears in the upper-right of the screen in **OPERATION:MEM** mode.

The first line tells you if the coolant is **ON** or **OFF**.

The next line shows the position number of the optional Programmable Coolant Spigot (**P-COOL**). The positions are from 1 to 34. If the option is not installed, no position number appears.

In the coolant gauge, a black arrow shows the coolant level. Full is 1/1 and empty is 0/1. To avoid coolant flow problems, keep the coolant level above the red range. You can also see this gauge in **DIAGNOSTICS** mode under the **GAUGES** tab.

## Timers & Counters Display

### F2.24: Timers & Counters Display Example



| Timers And Counters |         |
|---------------------|---------|
| This Cycle:         | 0:00:00 |
| Last Cycle:         | 0:00:00 |
| Remaining           | 0:00:00 |
| M30 Counter #1:     | 0       |
| M30 Counter #2:     | 0       |
| Loops Remaining:    | 0       |

The timer section of this display gives information about cycle times (This Cycle, Last Cycle, and Remaining).

The counter section has two M30 counters and a Loops Remaining display.

- M30 Counter #1: and M30 Counter #2: every time a program reaches an **M30** command, the counters increase by one. If Setting 118 is on, the counters also increment every time a program reaches an **M99** command.
- If you have macros, you can clear or change M30 Counter #1 with #3901 and M30 Counter #2 with #3902 (#3901=0).
- Refer to page **47** for information on how to reset the timers and counters.
- Loops Remaining: shows the number of subprogram loops remaining to complete the current cycle.

## Alarms and Messages Display

Use this display to learn more about machine alarms when they occur, view your machine's entire alarm history, look up definitions of alarms that can occur, view created messages, and show the keystroke history.

Press **[ALARMS]**, then select a display tab:

- The **ACTIVE ALARM** tab shows the alarms that currently affect machine operation. Use **[PAGE UP]** and **[PAGE DOWN]** to see the other active alarms.
- The **MESSAGES** tab shows the messages page. The text you put on this page stays there when you power the machine off. You can use this to leave messages and information for the next machine operator, etc.
- The **ALARM HISTORY** tab shows a list of the alarms that have recently affected machine operation. You can also search for an alarm number or alarm text. To do this type in the alarm number or the desired text and press **[F1]**.
- The **ALARM VIEWER** tab shows a detailed description of all the alarms. You can also search for an alarm number or alarm text. To do this type in the alarm number or the desired text and press **[F1]**.
- The **KEY HISTORY** tab shows up to the last 2000 keystrokes.

## Add Messages

You can save a message in the **MESSAGES** tab. Your message stays there until you remove it or change it, even when you turn the machine off.

1. Press **[ALARMS]**, select the **MESSAGES** tab, and press the **[DOWN]** cursor arrow key.
2. Type your message.

Press **[CANCEL]** to backspace and delete. Press **[DELETE]** to delete an entire line. Press **[ERASE PROGRAM]** to delete the entire message.

## System Status Bar

The System Status Bar is the read-only section of the screen located in the bottom, center. It displays messages for the user about actions they have taken.

# Position Display

The Position display shows the current axis position relative to four reference points (Work, Distance-to-go, Machine, and Operator). In any mode, press **[POSITION]** and use cursor keys to access the different reference points displayed in tabs. The last tab display shows all the reference points on the same screen.

## T2.19: Axis Position Reference Points

| Coordinate Display | Function                                                                                                                                                                                                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WORK (G54)         | This tab displays the axis positions relative to part zero. On power-up, this position uses work offset G54 automatically. It displays the axis positions relative to the most recently-used work offset.                                                                         |
| DIST TO GO         | This tab displays the distance remaining before the axes reach their commanded position. When in <b>SETUP : JOG</b> mode, you can use this position display to show a distance moved. Change modes (MEM, MDI) and then switch back to <b>SETUP : JOG</b> mode to zero this value. |
| MACHINE            | This tab displays the axis positions relative to machine zero.                                                                                                                                                                                                                    |
| OPERATOR           | This tab shows the distance you have jogged the axes. This does not necessarily represent the actual distance the axis is from machine zero, except when the machine is first powered on.                                                                                         |
| ALL                | This tab displays all reference points on the same screen.                                                                                                                                                                                                                        |

## Axis Display Selection

You can add or remove axes in the Positions displays. While a **Positions** display tab is active, press **[ALTER]**. The axis display selection window comes in from the right side of the screen.

**F2.25:** Axis Display Selector



Use the cursor arrow keys to highlight an axis, and press **[ENTER]** to toggle it on and off for display. The positions display will show axes that have a check mark. Press **[ALTER]** to close the axis display selector.



**NOTE:** You can display a maximum of (5) axes.

## Input Bar

**F2.26:** Input Bar



The input bar is the data entry section located in the bottom-left corner of the screen. This is where your input appears as you type it.

# Special Symbol Input

Some special symbols are not on the keypad.

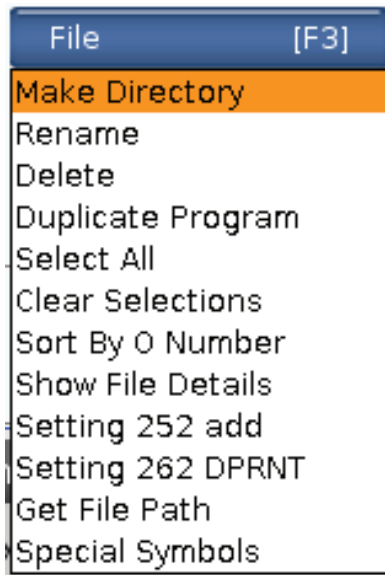
T2.20: Special Symbols

| Symbol | Name                  |
|--------|-----------------------|
| –      | underscore            |
| ^      | caret                 |
| ~      | tilde                 |
| {      | open curly brackets   |
| }      | closed curly brackets |
| \      | backslash             |
|        | pipe                  |
| <      | less than             |
| >      | greater than          |

Do these steps to enter special symbols:

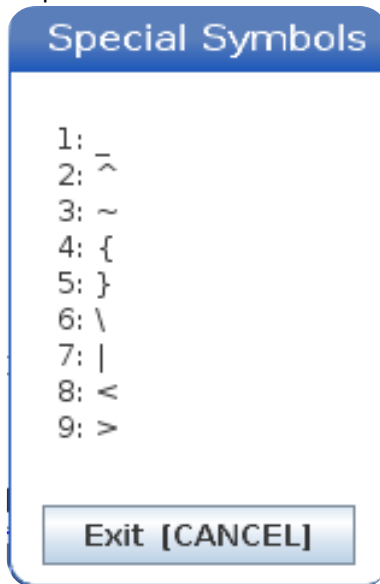
1. Press **[LIST PROGRAMS]** and select a storage device.
2. Press **[F3]**.

The **[FILE]** drop-down menu shows:



3. Select **Special Symbols** and press **[ENTER]**.

The **SPECIAL SYMBOLS** pick list shows:



4. Enter a number to copy the associated symbol to the **INPUT**: bar.

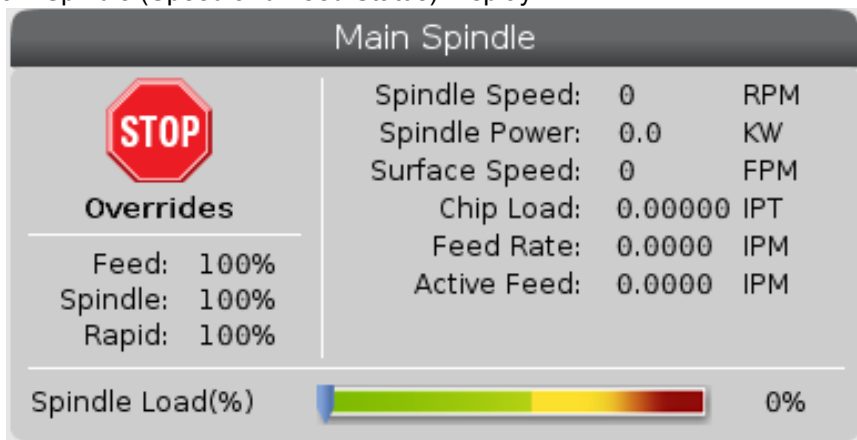
For example, to change a directory's name to **MY\_DIRECTORY**:

1. Highlight the directory with the name that you want to change.
2. Type **MY**.

3. Press **[F3]**.
4. Select **SPECIAL SYMBOLS** and press **[ENTER]**.
5. Press **[1]**.
6. Type **DIRECTORY**.
7. Press **[F3]**.
8. Select **RENAME** and press **[ENTER]**.

## Main Spindle Display

**F2.27:** Main Spindle (Speed and Feed Status) Display



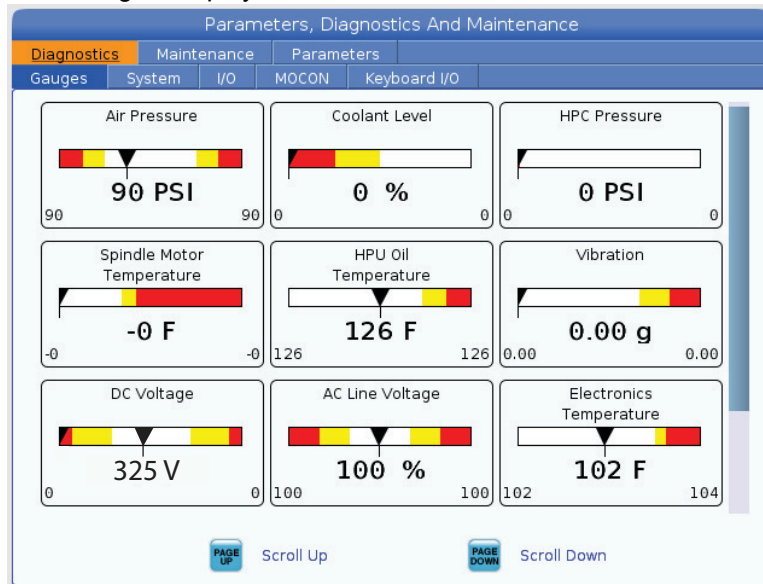
The first column of this display gives you information about feedrate, spindle, and rapid overrides.

The second column displays the current spindle speed in rpm and spindle load in kW. The spindle load value reflects the actual spindle power to the tool. The next values presented are linked: surface speed of the rotating tool in fpm, the actual chip load in in/tth, and the programmed feed rate in in/min. Active feed rate displays the actual feed rate including any manual overrides.

The spindle load meter indicates the spindle load as a percentage of motor capacity.

# Gauge Display

## F2.28: Diagnostic Gauges Display



This display quickly shows you information about various machine states, including fluid pressure, voltages, and component temperatures. Press the **[PAGE DOWN]** button to scroll to see more gauges.

Setting 9 determines the units that the gauges use for fluid pressure and temperature. If Setting 9 has a value of **INCH**, the gauges display air pressure in psi and temperature in degrees Fahrenheit. If Setting 9 has a value of **MM**, the gauges display fluid pressure in bar and temperature in degrees Celsius.

## Machine Activation Display

Contact your Haas Factory Outlet (HFO) for help with machine activation. Be prepared to give the HFO representative the information on this screen (Serial Number, Mac Address, Software Version, Activation Code).

## Diagnostics Display

This display shows information about your machine's configuration. Haas Service may ask you for some of the information from this screen when you call. You can also find information about machine run time, bill time, the number of tool changes, the number of power cycles, and total power-on time.

## Lubrication Tests Display

Haas Service technicians use this display to test your machine's lubrication system. Haas Service may also ask you to do these tests yourself. To avoid over-lubrication, you should not do these tests unless Haas Service directs you to do so.

### 2.2.5 Screen Capture

The control can capture and save an image of the current screen to an attached USB device or the User Data memory.

1. Press **[SHIFT]**.
2. Press **[F1]**.



#### NOTE:

*The control uses the default filename **snapshot#.png**. The # starts from 0 and increments each time you capture a screen. This counter resets at power off. Screen captures that you take after a power cycle overwrite previous screen captures that have the same filename on the User Data memory.*

The control saves the screen capture to your USB device or control memory. The message *Snapshot saved to USB* or *Snapshot saved to User Data* appears when the process finishes.

### 2.2.6 Error Report

The control can generate an error report that saves the state of the machine that is used for analysis. This is useful when helping the HFO troubleshoot an intermittent problem.

1. Press **[SHIFT]**.
2. Press **[F3]**.



#### NOTE:

*Be sure to always generate the error report with the alarm or the error is active.*

The control saves the error report to your USB device or control memory. The error report is a zip file that includes a screen capture, the active program, and other information used for diagnostics. Generate this error report when an error or an alarm occurs. E-mail the error report to your local Haas Factory Outlet.